

Self-Pruning Neural Network on CIFAR-10

S KISHOR

Data Science and Business Systems
SRM Institute of Science and Technology

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Advanced Deep Learning — Self-Pruning via Learnable Sigmoid Gates

1 Effect of L1 Regularisation on Sigmoid Gates

In this architecture, each weight w_{ij} is associated with a learnable gating parameter g_{ij} . The effective weight used during forward propagation is defined as:

$$W_{\text{eff}} = W \odot \sigma(4G)$$

where $\sigma(\cdot)$ denotes the sigmoid activation, ensuring all gate values lie within $(0, 1)$. The overall loss function combines task performance with sparsity regularisation:

$$L_{\text{total}} = \text{CrossEntropy}(\text{logits}, y) + \lambda \sum_{i,j} \sigma(4g_{ij})$$

Sparsity Induction Mechanism

The regularisation term applies a uniform penalty across all gates. During optimisation, the model reduces this penalty by shrinking gate activations, particularly for weights that contribute minimally to prediction accuracy.

If disabling a connection (i.e., pushing its gate towards zero) does not significantly increase the classification loss, the optimiser suppresses that connection. Conversely, weights critical for performance retain higher gate values, as pruning them would incur a larger loss penalty.

This interaction leads to a bimodal distribution of gate values:

- A dominant group near zero (inactive/pruned connections)
- A smaller group with higher values (active/important connections)

Mechanism	Outcome
Penalty on $\sum \sigma(4g)$	Encourages reduction of gate activations
$g \rightarrow -\infty$	$\sigma(4g) \rightarrow 0$, effectively disabling the weight
Retention of important weights	Prevents accuracy degradation
Scaling factor ($\times 4$)	Sharpens transition between active and inactive states

2 Experimental Results

λ	Test Accuracy	Sparsity (%)
1×10^{-5}	63.6%	5.8%
5×10^{-5}	63.6%	7.4%
1×10^{-4} (Best)	63.4%	97.4%

Discussion

- For $\lambda = 1 \times 10^{-5}$, the regularisation strength is insufficient to meaningfully suppress weights, resulting in minimal sparsity.
- Increasing to $\lambda = 5 \times 10^{-5}$ produces only a slight improvement, indicating that most connections are still preserved due to their contribution to task performance.
- At $\lambda = 1 \times 10^{-4}$, a significant transition occurs: over 97% of the weights are effectively pruned, while test accuracy decreases only marginally (by approximately 0.2%). This suggests that the model successfully isolates a compact and highly informative subset of parameters.

3 Gate Value Distribution Analysis

Best Configuration: $\lambda = 1 \times 10^{-4}$

The gate distribution corresponding to the optimal model exhibits a clear bimodal structure:

- **Peak near zero:** A large concentration of gates collapses towards zero, indicating that most connections have been pruned.
- **Secondary cluster:** A smaller group of gates remains at higher values, representing essential connections required for maintaining predictive accuracy.

This distribution confirms that the network has effectively learned a sparse subnetwork without significant loss in performance.

4 Design Considerations

Component	Purpose
Gate initialisation ($\mathcal{N}(0, 1) - 2$)	Biases gates towards initially closed states
Scaling factor ($\times 4$)	Produces sharper gating behaviour
Batch Normalisation	Stabilises activations under dynamic pruning
Cosine Annealing Scheduler	Enables gradual and stable convergence
Evaluation sigmoid ($\times 10$)	Provides clearer separation of active/inactive gates

5 Conclusion

This study demonstrates that incorporating learnable sigmoid gates with L1 regularisation enables effective self-pruning in neural networks. With an appropriate choice of regularisation strength, the model achieves substantial parameter reduction while preserving predictive performance.

The results highlight that a large proportion of network parameters are redundant, and structured sparsification techniques can uncover compact subnetworks that retain most of the original model’s capability.